

JEE MAIN REVISION MASTER: PROBABILITY 2026

5 QUESTIONS 8.33% WEIGHTAGE ↓16.7%

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 44 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Conditional Probability & Bayes' Theorem	24 PYQs	Conditional & Bayes
Very High	Probability Distribution & Binomial Distribution	20 PYQs	Random Variables

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Conditional Probability & Bayes' Theorem	$P(A B) = \frac{P(A \cap B)}{P(B)}$, $P(A \cap B) = P(A) \cdot P(B)$ (independent), $P(E_i A) = \frac{P(E_i)P(A E_i)}{\sum P(E_j)P(A E_j)}$
Probability Distribution & Binomial Distribution	$\text{Mean } \mu = \sum x_i p_i$, $\text{Var } \sigma^2 = \sum x_i^2 p_i - \mu^2$, $\text{Binomial: } P(X = r) = {}^n C_r p^r q^{n-r}$, $\text{Mean} = np$, $\text{Var} = npq$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Conditional Probability & Bayes' Theorem (Very High Yield — 24 PYQs)

Core Principle: Multiplication rule of probability, test for statistical independence, law of total probability, and inverse probability via Bayes' theorem.

■ **Caution / JEE Pitfall:** Mutually exclusive events ($A \cap B = \emptyset$ implies $P(A \cap B) = 0$) CANNOT be independent if both have non-zero probabilities!

■ **Speed Trick:** For independent events: $P(A \cup B) = 1 - P(A')P(B')$.

Representative Source PYQs:

- [2026 Shift] Q#2 (MCQ): "The probability that a man and his wife live after 20 years are $(1/4)$ and $(1/3)$ respectively. The probability that neither..." [Ans: D]
- [2026 Shift] Q#4 (MCQ): "Probability of occurrence of an event A is $(1/2)$ and that of B is $(3/10)$. If A and B are mutually exclusive, then the proba..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "A random experiment has five outcomes w_1, w_2, w_3, w_4 and w_5 . The probabilities of the occurrence of the outcomes $w_1, w_2, \dots$ " [Ans: N/A]

3.2 Probability Distribution & Binomial Distribution (Very High Yield — 20 PYQs)

Core Principle: Discrete probability distributions, expected value $E(X)$, variance, and Binomial Bernoulli trials ($q = 1 - p$).

■ **Caution / JEE Pitfall:** In a Binomial distribution, the variance (npq) is ALWAYS strictly less than the mean (np) because $q < 1$.

■ **Speed Trick:** Variance formula shortcut: $\text{Var}(X) = E(X^2) - [E(X)]^2$.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "Recent studies suggest that 12% of the world population is left handed. Depending on parents' hand usage, the chances of having left handed..." **[Ans: N/A]**
- **[2026 Shift] Q#1 (MCQ):** "Probability of at least one of the events A and B occur is 0.6. If A and B occur simultaneously with probability 0.2, then $P(\bar{A})$..." **[Ans: N/A]**
- **[2026 Shift] Q#3 (MCQ):** "Probability of obtaining an even prime number on each die when a pair of dice is rolled is..." **[Ans: D]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Recent studies suggest that 12% of the world population is left handed. Depending on parents' hand usage, the chances of having left handed children are as follows: A: Both parents are left handed, chances of having left handed children = 24% B: Both parents are right handed, chances of having left handed children = 9% C: Father left handed and mother right handed, chances of having left handed children = 17% D: Father right handed and mother left handed, chances of having left handed children = 22% Given $P(A)=P(B)=P(C)=P(D)=1/4$ and L denotes child is left handed. What is the probability that $P(A|L)$?

- (A) (17 / 100)
- (B) (19 / 25)
- (C) (1 / 3)
- (D) (2 / 3)

Correct Answer: None

Solution Analysis: We need to find $P(A|L)$, which means: Probability that both parents are left handed, given that the child is left handed. Using Bayes' theorem, $P(A|L) = (P(A)P(L|A) / P(L))$ Here, $P(A)=P(B)=P(C)=P(D)=(1/4)$ Also, $P(L|A)=24\%=(24/100)$ Now find $P(L)$ using the total probability formula:

$P(L)=P(A)P(L|A)+P(B)P(L|B)+P(C)P(L|C)+P(D)P(L|D)$ Substitute the values: $P(L)=(1/4) \times (24/100)+(1/4) \times (9/100)+(1/4) \times (17/100)+(1/4) \times (22/100)$ Since $(1/4)$ is common, $P(L)=(1/4)((24+9+17+22/100))$ $P(L)$

Benchmark Problem 2: KCET 2026 (Q#2)

The probability that a man and his wife live after 20 years are $(1/4)$ and $(1/3)$ respectively. The probability that neither the man nor his wife live after 20 years is

- (A) $(3/4)$
- (B) $(1/12)$
- (C) $(7/12)$
- (D) $(1/2)$ [Correct]

Correct Answer: D

Solution Analysis: Let the probability that the man lives after 20 years be $P(M)=(1/4)$ and the probability that his wife lives after 20 years be $P(W)=(1/3)$. So, the probability that the man does not live after 20 years is $1-(1/4)=(3/4)$ and the probability that the wife does not live after 20 years is $1-(1/3)=(2/3)$. Assuming their survival is independent, the probability that neither lives after 20 years is $(3/4) \times (2/3)=(1/2)$. So the correct answer is $(1/2)$ Option D